

SIMATS ENGINEERING

ASSESSMENT TOOL 2

Case study

COURSE NAME: Computer Networks

COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO3: *Analyze the performance of core network protocols such as TCP, UDP, HTTP, and DNS under varying conditions*

Assessment Tool Weightage: 40%

Case study

QUESTIONS:4

Total Marks:40

1. A multinational IT company uses a cloud-based video conferencing platform for daily client meetings. During peak business hours, employees experience frequent voice distortion, frozen video frames, and delayed communication. Network monitoring reports indicate an **8% packet loss**, **180 ms jitter**, and fluctuating bandwidth utilization. The IT administrator suspects that the issue is related to the transport protocol or application-level buffering strategy.

Questions

1. Analyze whether the problem is primarily caused by the **transport protocol (TCP/UDP)** or the **application layer**.
2. Justify your answer based on the communication requirements of real-time multimedia applications.
3. Recommend suitable protocol-level or application-level improvements to enhance conferencing quality.

ANS:

1. Analyze whether the problem is primarily caused by TCP/UDP or the application layer:

The problem is not caused by only TCP/UDP. It is mainly a network-quality problem affecting the transport and application layers, with the application's buffering and congestion-handling strategy playing an important role.

- 8% packet loss is very high for video conferencing and causes missing audio/video packets.
- 180 ms jitter is also very high and causes packets to arrive at irregular intervals.
- Fluctuating bandwidth can cause congestion and inconsistent data delivery.
- If the application uses TCP, retransmission of lost packets can create additional delay and video freezing.
- Most modern real-time conferencing systems prefer UDP-based transport because avoiding retransmission delays is more important than guaranteeing delivery of every packet.
- Therefore, if the platform already uses UDP, the application should be examined for jitter buffering, adaptive bitrate, congestion control, and packet-loss concealment.

Conclusion: The immediate symptoms are primarily due to poor network conditions, while the transport protocol and application-level buffering determine how severely users experience those conditions.

2. Justification based on real-time multimedia requirements:

Real-time multimedia applications such as video conferencing require:

1. Low latency – Voice and video must arrive quickly for natural conversation.
2. Low jitter – Packets should arrive at nearly regular intervals.
3. Low packet loss – Excessive loss produces distorted audio, missing frames, and frozen video.
4. Stable bandwidth – Video requires continuous bandwidth, especially at higher resolutions.
5. Fast recovery – The system should handle lost packets without causing long delays.

UDP is generally more suitable than TCP for real-time communication because UDP does not wait for lost packets to be retransmitted. In TCP, retransmission and ordered delivery can cause head-of-line blocking, making already-delayed video/audio even more delayed.

For example:

TCP:

Packet lost → retransmission → waiting → increased delay → frozen/delayed media

UDP:

Packet lost → continue with available packets → application conceals the loss → lower delay

Thus, for interactive voice/video, timeliness is usually more important than perfect delivery of every packet.

3. Recommended protocol-level and application-level improvements:

A. Protocol-level improvements:

- Prefer UDP-based real-time transport instead of TCP for audio/video media.
- Use RTP/RTCP for real-time media transmission and monitoring.
- Implement congestion control to reduce the sending rate when bandwidth decreases.
- Use forward error correction (FEC) to recover from some packet losses without retransmission.
- Use selective retransmission/NACK only when the missing packet can still be useful within the application's playback deadline.
- Apply QoS prioritization so voice and video packets receive higher priority than less time-sensitive traffic.
- Use traffic shaping and bandwidth management to reduce congestion during peak hours.

B. Application-level improvements:

- Implement an adaptive jitter buffer to compensate for variations in packet arrival time.
- Use adaptive bitrate (ABR) so video quality automatically decreases when bandwidth becomes insufficient.
- Reduce video resolution or frame rate during congestion rather than allowing the video to freeze.
- Use packet-loss concealment for audio so missing packets produce less noticeable distortion.
- Dynamically adjust the audio/video encoding rate according to network conditions.
- Monitor packet loss, jitter, latency, and available bandwidth and automatically optimize media quality.
- Use efficient codecs such as Opus for audio and modern video codecs suitable for real-time communication.

Final conclusion:

The 8% packet loss and 180 ms jitter indicate severe network instability. UDP is generally preferable for real-time conferencing because it minimizes delay, but simply switching protocols will not solve the problem. The best solution is a combination of UDP/RTP, congestion control, QoS, adaptive bitrate, adaptive jitter buffering, FEC, and packet-loss concealment. This allows the conferencing application to maintain acceptable audio and video quality even when network conditions fluctuate.

2. A multinational software company hosts its web services across multiple continents. Employees frequently report long delays before the application homepage loads. Network analysis shows that the **average DNS lookup time is approximately 800 ms**, significantly affecting user experience. The organization plans to redesign its DNS infrastructure while ensuring continuous service availability during server failures.

Questions

1. Analyze the reasons behind the high DNS resolution time.
2. Propose a suitable DNS architecture using techniques such as **Anycast DNS**, **DNS pre-fetching**, and **TTL optimization** to reduce the lookup time below **50 ms**.
3. Evaluate the advantages and limitations of your proposed architecture in terms of performance, scalability, and availability.

ANS:

1. Reasons for High DNS Resolution Time:

The 800 ms DNS lookup time is very high. The main reasons are:

- DNS servers may be far away from the users.
- DNS servers may be overloaded during peak hours.
- DNS cache may have expired, causing repeated DNS lookups.
- Low TTL values can cause records to expire frequently.
- Network congestion can increase DNS response time.
- The company may not have enough DNS servers in different locations.

2. Proposed DNS Architecture:

The company can use Anycast DNS + DNS pre-fetching + TTL optimization.

Anycast DNS:

- Deploy DNS servers in different continents.
- Use the same Anycast IP address for all DNS servers.
- Users are automatically directed to a nearby DNS server.
- This reduces DNS response time.

DNS Pre-fetching:

- Frequently used DNS records are refreshed **before they expire**.
- This reduces cache misses.
- Users can get DNS responses faster.

TTL Optimization:

- Use a suitable TTL value for DNS records.
- Longer TTL for records that do not change frequently.
- This keeps records in cache for longer and reduces DNS queries.

Fault Tolerance:

- Keep multiple DNS servers in different regions.
- If one server fails, another server can handle the requests.
- This provides continuous service availability.

Target: With these techniques, cached DNS responses can be designed to respond in **less than 50 ms** for users with good network connectivity.

3. Advantages and Limitations:

Aspect	Advantages	Limitations
Performance	Faster DNS responses	Initial uncached lookup may still take time
Scalability	Requests are distributed across many servers	More servers are required
Availability	Service continues if one server fails	Requires proper configuration
Anycast DNS	Directs users to nearby DNS servers	More complex to manage
Pre-fetching	Reduces cache misses	Creates some additional DNS traffic
TTL Optimization	Reduces repeated DNS lookups	Long TTL can delay DNS changes

Final Conclusion:

The company should use Anycast DNS, DNS pre-fetching, optimized TTL values, and multiple backup DNS servers. This will reduce DNS lookup time, improve scalability, and provide high availability even if a DNS server fails.

3. A cloud service provider delivers applications to millions of users worldwide. During promotional events, users experience intermittent delays in accessing cloud-hosted services. Investigation reveals that DNS queries are taking nearly **800 ms** because all requests are directed to a single primary DNS server. The company wants a highly available DNS solution capable of handling global traffic efficiently.

Questions:

- 4. Examine the impact of centralized DNS architecture on application performance.
- 5. Design a distributed DNS solution that minimizes lookup latency while maintaining high availability.
- 6. Analyze how **TTL values**, **Anycast routing**, and **DNS caching** influence system performance and fault tolerance.

ANS:

1. Impact of Centralized DNS Architecture:

A centralized DNS architecture uses one main DNS server to handle all DNS requests.

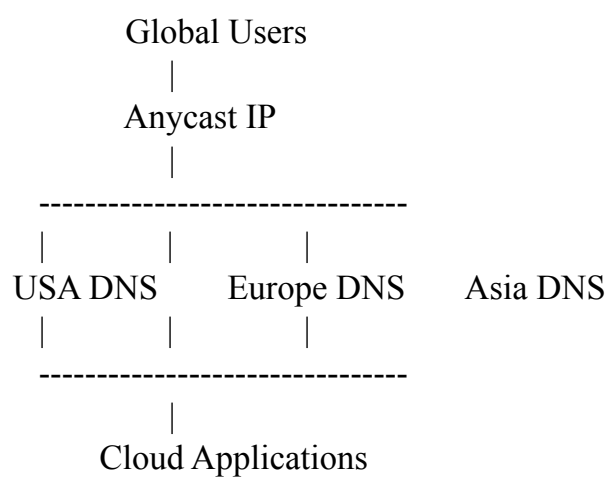
The problems are:

- **High latency:** All users must contact the same server, causing the **800 ms delay**.
- **Server overload:** During promotional events, millions of requests can overload the server.
- **Single point of failure:** If the primary DNS server fails, users may not be able to access the applications.
- **Poor global performance:** Users far away from the server experience higher network delay.
- **Slow application loading:** DNS must complete before the application connection starts.

2. Distributed DNS Solution:

The company should use multiple DNS servers in different geographical locations.

Architecture:



Working:

- 1. Deploy DNS servers in different regions.
 - 2. Give them a common Anycast IP address.
 - 3. Users are automatically directed to a nearby DNS server.
 - 4. DNS responses are stored in caches.
 - 5. Multiple DNS servers provide backup if one server fails.
 - 6. Load is distributed between servers.
- This reduces DNS lookup latency and prevents one server from becoming overloaded.

3. Effect of TTL, Anycast Routing and DNS Caching:

A. TTL (Time To Live):

TTL determines how long a DNS record stays in a cache.

- **Higher TTL:** Fewer DNS queries and better performance.
- **Lower TTL:** DNS changes are updated faster but create more queries.

Best approach: Use a suitable TTL based on how frequently the DNS records change.

B. Anycast Routing:

Anycast allows multiple DNS servers to use the same IP address.

- Users are directed to a nearby DNS server.
- Reduces network delay.
- Distributes traffic between DNS servers.
- If one server fails, traffic can be routed to another available server.

Result: Better performance and high availability.

C. DNS Caching:

DNS caching stores previously resolved DNS records.

For example:

First request → DNS lookup → Store in cache
Next request → Get answer from cache → Faster

Benefits:

- Reduces DNS lookup time.
- Reduces load on DNS servers.
- Improves application loading speed.
- Allows the system to continue serving cached records during some DNS server failures.

Final Conclusion:

The company should replace the single centralized DNS server with a distributed Anycast DNS architecture using multiple global DNS servers, proper TTL values, and DNS caching.

This solution will:

- Reduce DNS latency
- Handle millions of users
- Prevent server overload
- Improve fault tolerance
- Provide high availability
- Improve overall application performance

4. A financial institution operates an electronic stock trading platform where thousands of buy and sell orders are submitted every second. During market opening hours, the **average order acknowledgment latency increases from 1 ms to 40 ms**, resulting in delayed trade confirmations and customer dissatisfaction. Network engineers observe increased TCP retransmissions, longer connection establishment times, and excessive buffering.

Questions

1. Analyze three possible TCP-related factors responsible for the latency increase, considering **flow control, Nagle's algorithm, and SYN queue limitations**.
2. Explain how each factor affects transaction processing in high-frequency systems
3. Recommend appropriate TCP configuration changes to reduce latency and improve transaction reliability.

ANS:

1. Three TCP-related factors causing latency:

A. Flow Control:

TCP uses a receive window to control how much data can be sent before receiving an acknowledgment.

During market opening, thousands of orders are sent at the same time. If the receiver's buffer becomes full:

- TCP reduces the sending rate.
- Data waits in the buffer.
- More waiting causes higher latency.
- Order acknowledgments are delayed.

Effect: The latency can increase from **1 ms to 40 ms**.

B. Nagle's Algorithm:

Nagle's algorithm combines small TCP packets into larger packets to reduce network overhead.

For stock trading, orders and acknowledgments are often small messages.

Nagle's algorithm may:

- Hold a small packet temporarily.
- Wait for an acknowledgment or more data.
- Add extra delay to each transaction.

Effect: It can be harmful to high-frequency trading where very low latency is required.

Solution: Disable Nagle's algorithm using TCP_NODELAY for latency-sensitive connections.

C. SYN Queue Limitations:

When many clients connect at the same time, the server receives a large number of TCP SYN requests.

If the SYN queue becomes full:

- New connections may be delayed or dropped.
- Clients retransmit SYN packets.
- Connection establishment takes longer.
- Order processing is delayed.

Effect: During market opening, the sudden traffic increase can cause significant connection delays.

2. Effect on High-Frequency Transaction Processing:

TCP Factor	Effect
Flow Control	Receiver buffer fills → sending slows → orders wait
Nagle's Algorithm	Small packets are delayed → order acknowledgment takes longer
SYN Queue	Too many connection requests → connection establishment becomes slow
TCP Retransmissions	Lost packets are retransmitted → additional delay

In a high-frequency trading system, even a few milliseconds of additional delay can affect order execution and trade confirmation.

3. Recommended TCP Configuration Changes:

1. Disable Nagle's Algorithm:

Use:

TCP_NODELAY

This allows small trading messages to be sent immediately instead of waiting for packet aggregation.

2. Increase TCP Buffer Sizes:

Configure suitable send and receive buffers so that temporary traffic bursts do not immediately cause flow-control delays.

3. Increase SYN Queue Size:

Increase the server's SYN backlog/queue to handle large numbers of simultaneous connections.

4. Use Persistent TCP Connections:

Instead of creating a new TCP connection for every order, maintain long-lived connections.

This avoids repeated TCP handshakes and reduces connection establishment time.

5. Reduce Packet Loss:

Since TCP retransmissions are increasing, improve network reliability and reduce congestion.

6. Monitor TCP Performance:

Monitor:

- Retransmissions
- RTT
- Send/receive window
- SYN queue usage
- Buffer utilization

Final Conclusion:

The increase from 1 ms to 40 ms can be caused by flow-control delays, Nagle's algorithm, and SYN queue limitations, along with increased retransmissions. For a high-frequency trading platform, the best approach is to disable Nagle's algorithm, optimize TCP buffers, increase the SYN backlog, use persistent connections, and reduce packet loss. These changes can significantly reduce latency while maintaining reliable order delivery.

RUBRICS FOR CALCULATION

Criteria	Weight ag e	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developi ng	Level 1 – Beginning
Understandin g of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understan di ng; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limite d applicati on of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Soluti on Desig n	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justificati on	Incorrect or no solution provided
Clarity	10	Clear, well- structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand